

Sydney Housing Price Prediction and Decision Support System

SIT307 8.1 Distinction Task

Suburbs analysed: Newtown, Castle Hill, Cronulla

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Part 1: Problem Definition and Data Collection

Suburb Selection and Motivation

Three Sydney suburbs were chosen to represent genuinely different lifestyle and commute profiles, rather than a simple cheap/mid/expensive price gradient. **Newtown** is an inner-west, high-density suburb roughly 4km from the CBD, dominated by units and studios, with strong gentrification and rental demand driven by its walkability and train-line access. **Castle Hill** sits in the Hills District, around 32km from the CBD, and contains a mix of units near the newer Sydney Metro station alongside larger detached family houses on much bigger blocks of land. **Cronulla** is a beachside suburb about 28km from the CBD where houses and semi-detached properties dominate and sale prices carry a clear lifestyle premium tied to beach proximity. Each suburb attracts a different kind of buyer: renters and young professionals chasing walkability in Newtown, families trading commute time for space in Castle Hill, and lifestyle buyers paying for beach access in Cronulla. That spread is exactly what a decision-support tool needs to be useful across more than one kind of market.

Problem Statement

The task is framed as a supervised regression problem: predict a property's sale price (a continuous target) from its observable characteristics, in order to support valuation decisions across three structurally different markets.

Data Collection

97 sold properties were manually collected from Domain's "sold listings" pages (no scraping, per the assignment requirement), covering sales from late 2025 through September 2026: 33 in Newtown, 31 in Castle Hill, and 33 in Cronulla. Core fields (price, address, date, property type, bedrooms, bathrooms, car spaces) were recorded for every property. `land_size_sqm` was only available for 45 of the 97 rows, since strata properties (units, studios) typically don't list a separate land size on Domain. Optional fields (building size, year built, school zone, agent description) were not consistently listed and were excluded from modelling entirely rather than partially filled in, to avoid introducing noise. Distance to the CBD, nearest train station, and the beach were recorded as fixed per-suburb constants rather than looked up per property, which is a deliberate simplification made to keep data collection feasible within the time available — it can't capture, for example, that a Cronulla property directly on the esplanade is meaningfully different from one 2km inland. The dataset also only reflects listed properties that successfully sold, so it excludes off-market sales and auctions that were passed in, which is a source of potential selection bias worth flagging.

Part 2: Data Understanding and Feature Engineering

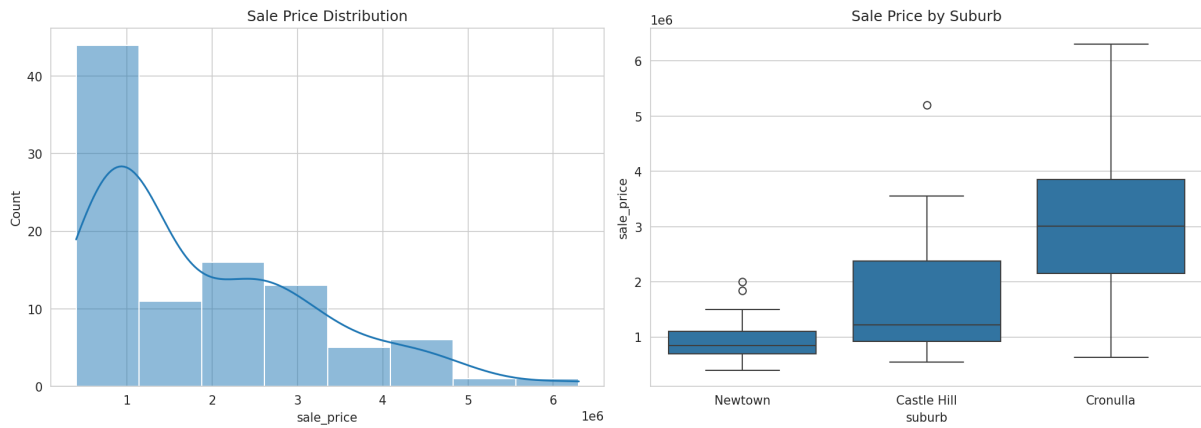


Figure 1: Overall sale price distribution (left) and sale price by suburb (right).

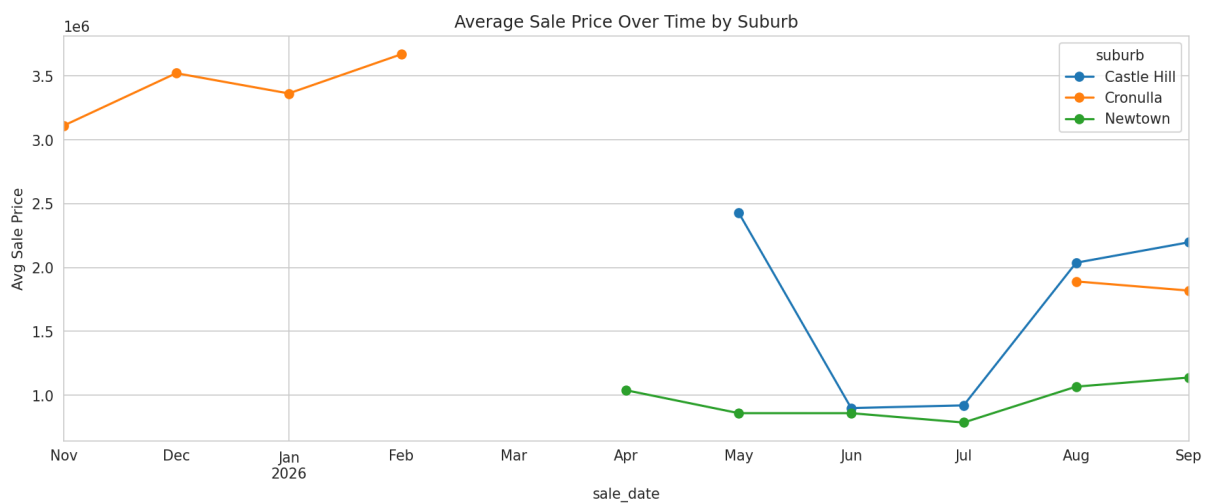


Figure 2: Average sale price over time, by suburb.

The price distribution is heavily right-skewed overall, which is expected once three very different markets are combined into one dataset — Cronulla's high-end houses (several sales above \$4M) sit alongside Newtown studios selling for under \$450,000. The by-suburb boxplot makes the market separation clear: Newtown's spread is tight and low relative to Castle Hill and Cronulla, Castle Hill spans a wide range because it mixes cheaper units with large family houses, and Cronulla has both the highest median and the widest spread of the three, driven by its premium beachfront houses.

Outliers

Applying the IQR method per suburb (rather than globally, since a "normal" Cronulla price would be a huge outlier in Newtown) flagged three properties: two Newtown houses that sold well above the unit-dominated local market (\$1.84M and \$2.0M, against a suburb where most sales are units under \$1.3M), and one exceptional Castle Hill sale at \$5.2M on a 1,284 sqm block, far above the rest of the suburb. These aren't data errors — they're genuinely higher-value properties within their local markets and were kept in the dataset, since excluding them would bias the model against correctly pricing large or premium properties.

Predicted Top 3 Features (Before Engineering)

Before engineering or checking correlations, the hypothesis was that **suburb** would be the dominant driver (it bundles location, lifestyle premium and land scarcity into one variable), followed by **bedrooms** as a proxy for overall property size and type, and **land_size_sqm** mattering heavily specifically within the Castle Hill and Cronulla house markets.

Feature Correlations and Reflection

Feature	Correlation with sale_price
bedrooms	0.795
land_size_sqm	0.739
bathrooms	0.706
car_spaces	0.664
distance_to_cbd_km	0.471
distance_to_train_km	0.213
distance_to_beach_km	-0.193

This confirms bedrooms and land size as strong individual predictors, matching the hypothesis. Suburb doesn't appear in this table since it's categorical, but Figure 1's boxplot already shows it separates prices at least as strongly as any single numeric feature — if anything, the hypothesis undersold how dominant suburb turns out to be once combined with property type. Notably, distance to the beach shows a weak negative correlation (-0.19) rather than the strong effect expected: because it's a suburb-level constant, it's really acting as another encoding of suburb rather than a genuine within-suburb signal.

Part 3: Model Development and Evaluation

Three regression models were trained: Linear Regression, Random Forest, and Gradient Boosting. Before training, Linear Regression was expected to underperform since price doesn't scale linearly with features across such different property types. Random Forest was expected to do reasonably well given its ability to handle non-linear, categorical splits with a small dataset. Gradient Boosting was expected to fit best on raw performance, but with a real risk of overfitting given only 97 rows split across three suburbs — cross-validation was intended as the real test of that risk.

Results

Model	CV RMSE	Test RMSE	Test MAE	Test R ²
Linear Regression	\$553,922	\$552,647	\$371,827	0.811
Random Forest	\$572,787	\$419,195	\$312,037	0.891
Gradient Boosting	\$620,370	\$438,291	\$309,423	0.881

Critical Analysis

Random Forest achieved the lowest test RMSE (\$419,195) and highest R² (0.891), meaning it explains roughly 89% of the variance in sale price on unseen data. Gradient Boosting actually had the *worst* mean CV RMSE of the three models, despite a competitive test score — this is a real sign of instability rather than a fluke, and lines up with the pre-training concern that its extra complexity would struggle on a 97-row dataset. Linear Regression's CV and test RMSE are almost identical, which shows it isn't overfitting, but it simply lacks the capacity to capture the non-linear jump in price between, say, a Newtown studio and a Castle Hill house.

The pre-training predictions were partly right: Linear Regression did underperform as expected, and Gradient Boosting's overfitting risk did materialise in the CV numbers, even though its test RMSE looked fine on this particular train/test split. Random Forest's balance of CV stability and test accuracy makes it the strongest choice.

Recommended Model

Random Forest is recommended: it has the best test-set performance, and its cross-validation RMSE is close enough to its test RMSE to suggest the result isn't a product of a lucky train/test split, unlike Gradient Boosting's more volatile cross-validation scores.

Part 4: Investigating Prediction Failures

Address	Suburb	Type	Bed/Bath	Actual	Predicted	Abs. Error
16 Sturt Road	Cronulla	House	4/1	\$2,400,000	\$3,352,441	\$952,441
173 Ewos Parade	Cronulla	House	4/1	\$4,550,000	\$3,696,654	\$853,346
9b Alfred Avenue	Cronulla	House	4/3	\$3,200,000	\$3,813,220	\$613,220
4E Bate Bay Road	Cronulla	Semi-det.	3/3	\$3,210,000	\$2,671,841	\$538,159
95 Station St	Newtown	House	2/1	\$1,842,000	\$1,318,862	\$523,138

Case-by-Case Analysis

Four of the five worst predictions are Cronulla houses, and the fifth is a Newtown house — an unusual property type for a suburb the model has mostly learned as a unit/studio market. 16 Sturt Road and 9b Alfred Avenue were both *overpriced* by the model relative to their actual sale price, while 173 Ewos Parade, 4E Bate Bay Road, and 95 Station Street were all *underpriced*. This split in direction suggests the errors aren't a simple systematic bias but reflect genuine within-suburb variation the model can't see: it has no data on ocean views, street position, or renovation quality, so two Cronulla houses with identical bedroom/bathroom/land-size numbers can have very different real prices depending on whether one is beachfront and the other is a block back. 95 Station Street, Newtown is a similar story in reverse — the model has learned Newtown as a low-price unit-dominated suburb, so a full house there gets underpriced because the model hasn't seen enough Newtown houses to learn that segment properly.

Missing Information

The clearest gap is anything capturing *within-suburb* location quality — exact distance to the beach or CBD per property rather than a suburb-level constant, walk score, street type, or view aspect. Renovation status and condition were also never collected, despite being one of the biggest real-world price drivers for older housing stock in suburbs like Cronulla and Newtown. Agent description text was collected for none of the properties in this run, which could otherwise have offered a cheap way to capture qualitative signals like "ocean views" or "fully renovated" through simple keyword flags.

Harder-to-Model Property Types

High-end Cronulla houses are clearly the hardest segment: they show the widest spread in the boxplot and dominate the worst-prediction table, since their prices are driven heavily by qualitative lifestyle factors (views, proximity to the esplanade) that aren't in the dataset. Properties that are structurally atypical for their suburb — a house in a unit-dominated market like Newtown, or vice versa — are also harder to model, simply because there are fewer comparable training examples for the model to learn from.

Part 5: Final Deployment and Reflection

A Streamlit web app was built to let a user enter suburb, property type, bedrooms, bathrooms, car spaces, and land size, and receive a predicted sale price from the trained Random Forest model. The app was run locally (streamlit run app.py) and tested with three inputs spanning the dataset's range:

 **Sydney Housing Price Predictor**

Predicts sale price for Newtown, Castle Hill, and Cronulla properties based on a Gradient Boosting model trained on 97 manually collected sold listings.

Suburb
Cronulla

Property Type
House

Bedrooms
4

Bathrooms
2

Car Spaces
1

Land Size (sqm) — leave at 0 if unknown/apartment
600

Predict Price

Predicted Sale Price: \$2,615,924

Built for SIT307 8.1 Distinction Task — for educational purposes only, not real valuation advice.

Figure 3: App prediction for a Cronulla house (4 bed / 2 bath / 1 car / 600 sqm) — \$2,615,924.



Sydney Housing Price Predictor

Predicts sale price for Newtown, Castle Hill, and Cronulla properties based on a Gradient Boosting model trained on 97 manually collected sold listings.

Suburb	Car Spaces
Newtown	1
Property Type	Land Size (sqm) — leave at 0 if unknown/apartment
Unit	0
Bedrooms	
1	
Bathrooms	
1	
Predict Price	

Predicted Sale Price: \$1,029,788

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Figure 4: App prediction for a Newtown unit (1 bed / 1 bath / 1 car) — \$1,029,788.



Sydney Housing Price Predictor

Predicts sale price for Newtown, Castle Hill, and Cronulla properties based on a Gradient Boosting model trained on 97 manually collected sold listings.

Suburb	Car Spaces
Castle Hill	0
Property Type	Land Size (sqm) — leave at 0 if unknown/apartment
House	900
Bedrooms	
5	
Bathrooms	
3	
Predict Price	

Predicted Sale Price: \$1,952,985

Built for SIT307 8.1 Distinction Task — for educational purposes only, not real valuation advice.

Figure 5: App prediction for a Castle Hill house (5 bed / 3 bath / 0 car / 900 sqm) — \$1,952,985.

The three predictions land in a sensible order — the Newtown unit lowest, the Castle Hill house in the middle, and the Cronulla house highest — which is a reasonable sanity check that the model has genuinely learned the price differences between the three markets rather than just memorising an average.

Reflection

Manual data collection was by far the most time-consuming part of this task, and it directly shaped what the model could and couldn't learn: fields that were tedious to find (land size on strata listings, renovation status, exact distances) were the same fields that ended up mattering most for the properties the model got wrong. There's a clear trade-off between data collection effort and model accuracy — a small amount of extra time spent recording exact walking distance to the beach, or a simple "renovated: yes/no" flag, would likely have closed a meaningful chunk of the error seen in Part 4, more than switching to a fancier model would have. On fairness and bias, it's worth being upfront that this dataset only reflects properties that were successfully listed and sold through Domain within a roughly 10-month window; it excludes off-market sales, properties that were passed in at auction, and any seasonal effects outside that window, so the model shouldn't be treated as a general-purpose valuation tool beyond these three suburbs and this time period. With more time, the priorities would be: collecting per-property (not per-suburb) distance data, adding a renovation/condition flag, and collecting more Newtown house sales specifically, since that segment was under-represented relative to units and was where one of the biggest prediction errors showed up.

GenAI Acknowledgement

I used Claude to help scaffold this project — setting up the folder structure, the notebook skeleton for all five parts, and the boilerplate code for the modelling pipeline (train/test split, cross-validation loop, the Streamlit app template). I collected all 97 property records myself manually from Domain's sold listings, since that's a hard requirement of the task and something I needed to actually look at to write the discussion sections properly. I ran the notebook myself, reviewed the actual output numbers (the model comparison table, the worst-5 predictions, the correlations), and wrote the analysis and reflection in this report based on what those numbers actually showed, rather than pasting in generic commentary. I also tested the deployed app myself and took the screenshots above.